Sizing rooftop PV and batteries after energy efficiency: an optimisation framework for manufacturing SMEs

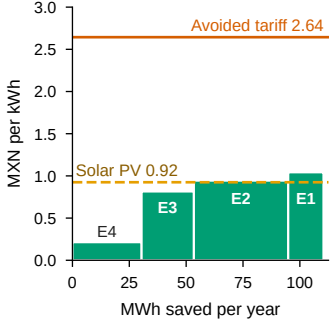
Data a typical SME already has

- 28 monthly bills
- Production records
- Walk-through audit
- Official tariffs, emission factor, fuels
- NASA POWER weather

Bills reproduced within 2.2%

1 Cut waste

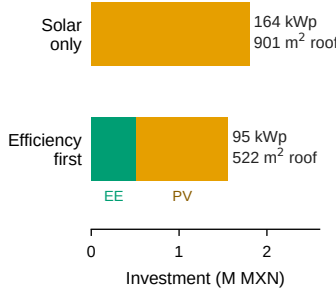
Cost of saved electricity



−23% grid electricity for 0.52 M MXN

2 Then size solar

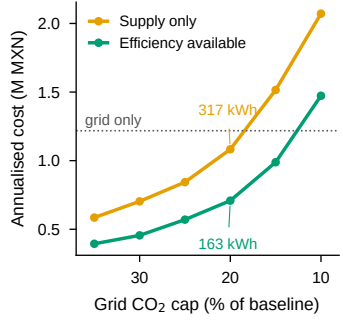
Same 54% cut in grid purchases



−13% capital, −42% roof, higher NPV in 97% of runs

3 Optimise deep cuts (urbs)

Least-cost system under CO_2 caps



Efficiency enters first: −29–35% cost, about half the storage

Net zero, security and resilience

- Annual net-zero electricity pays: 234 kWp after efficiency vs 303 kWp without
- If PV loses demand credit: solar NPV −69%, efficiency unaffected
- 111 tCO_2 and 850 GJ of imported gas avoided per year